CMP5366 Coursework Task Sheet 1
Data Analytics Pipeline: Design & Implementation

Your first task will be to **plan**, **design and implement** a **data management** and **data analytics** strategy for a specific problem domain. Your objective is to utilise data effectively to generate analytical insights and develop machine learning models. The details of your approach and design choices should be documented in your log report.

The purpose of this task is to contribute towards meeting the module learning outcomes:
"Plan and justify a pipeline and associated data storage system to automate the development process of a Machine Learning model."
"Implement a Machine Learning pipeline and associated data storage system using appropriate software."

Underlined are the five key parts of the learning outcomes, but what do they mean?

- Plan
  - You are expected to design a structured pipeline that outlines the key stages of data processing, storage, and model development. Your plan should be systematic, scalable, and suitable for automating the Machine Learning development process.
- Justify
  - Every design decision should be clearly explained and well-supported. Your justification should demonstrate why the chosen approach is the most suitable and to allow the reader to better understand your thought process when attempting the planning and design task.
- Data Storage System
  - The modelling process needs to consider not only how the data will be stored, but how it will be used, this should also be communicated appropriately.
- Automate the Development Process
  - The pipeline should be designed with automation in mind, minimizing manual intervention.
  - The modelling process must also consider that the stored data will be used to develop Machine Learning models and apply various analytics techniques. These considerations must also be communicated to the reader.
- Implement
  - You are expected to implement your designed pipeline using the tools and techniques taught in this module.

Note
The information contained within this document may be subject to further change to provide further clarification of aspects of the coursework, or to provide additional advice/guidance for the task.

Getting started

Exercise 1: Step 1.1

Identify **two candidate data sets**[1] that could be leveraged to develop a Machine Learning model to solve a supervised learning task (e.g. predicting flight delays). Ensure that these datasets have been previously used in scientific papers.

For each candidate source data set you should attempt to **discuss** and **justify** the following:

- **What** is it?
  - Describe the dataset and the type of data it represents (statistical data type/level of measurement).
  - e.g., what data does it represent?
- **Who** / what was it collected from?
  - Identify the original data sources. Where was the data originally sampled from?
    - e.g. Is it collected from particular patients at a particular set of hospitals?
- **Why** was it collected?
  - Explain the purpose of the dataset's existence.
  - e.g., why does this data exist in the first place?
- **Where** did you get it from?
  - Provide the source of the dataset (website, research paper, repository, API, etc.).
  - Where did you find it?
- **What** is its logical structure?
  - Describe the structural format of the dataset.
  - e.g., Is it a flat file, a collection of tables in a relational database, or structured differently?
  - Provide a Logical Diagram
    - What does it "look" like structurally? Document this
    - Show the **logical** structure of the input data using an appropriate diagram such as an **Entity-Relationship Diagram (ERD)**
- **How** is it currently stored?
  - e.g., Is it from a local database? A file? A web service?
  - e.g., What kind of database? How is the data currently physically stored using that database engine?
  - e.g., What kind of file (CSV, JSON, etc.)? How is data physically stored within that file on the file system?
- **How** could it be used to develop a machine learning model?
  - Explain how the dataset can be used for Machine Learning model development.
  - Define key features: Predictor variables and Target variables
    - What parts of the dataset could be leveraged to develop the model?
      - e.g. in a supervised learning task: what is the target variable and what are the predictor variables?
      - What is the **"Ground truth"**?
  - Has this dataset previously been used to develop Machine Learning models?
    - Explain and provide **reference**
    - If not, justify why?

[1] We recommend at least one of them be the same dataset you worked for Machine Learning & AI Module

Exercise 1: Step 1.2

From the **candidate datasets** identified in Step 1.1, select the **most suitable dataset** for developing a Machine Learning model. This dataset will be used later to plan a **data analytics pipeline** that will ingest, process, and utilise the data effectively.

- Discuss the reasoning behind your choice of data set as a group.
  - Ensure you precisely identify how you believe the source dataset could be leveraged for Machine Learning.

Warning!

For the particularly ambitious, you may wish to investigate how two or more dataset sources could be **combined from their different data sources into a single dataset**[2] to augment your dataset with additional data from another source.

A note of caution: you must be very careful when attempting to establish relationships between data from disparate sources as they may not necessarily be suitable to combine.

[2] We sometimes call this "Data Blending" when it is done for a specific use case, this can also be compared with the ETL process

Exercise 1: Step 2

The documentation of your pipeline should be accompanied by an appropriate plan for the storage of data within your analytics system. Therefore, this step requires you to document **how your chosen dataset will be stored** within the analytics system. Your discussion should provide a clear and well-justified plan for both the logical and physical structure of the data.

Note 1: To avoid redundancy, refer to information provided in earlier sections of your report instead of repeating details. If you have already discussed a particular aspect of the dataset (e.g., its origin or structure), simply reference that section rather than restating the same information.
Note 2: You should provide a detailed, well-justified discussion of each of the following components.

1. **Detail and justify** an appropriate logical and physical structure for the storage of data when it has been ingested.
   - Physical Storage Structure
     - Where will the ingested data be stored?
     - How will it be physically stored?
   - Logical Storage Structure
     - How will it be represented and structured logically to the Data Engineers and Data Scientists who work with it?
       - One Big Table?
       - Dimensional Model?
         - Star Schema, Snowflake Schema, etc.
       - Normalised Relational Model?
   - Provide comparison of Structures:
     - What are the <u>benefits and drawbacks</u> of your chosen structure?
     - What <u>alternative structures</u> could be used and What are the <u>benefits and drawbacks</u> of these alternative structures?
2. **Detail the process** of how data will be extracted, loaded (and possibly transformed) from the original data source before being stored for later use by the analytics system
   - e.g. ELT vs ETL

Exercise 1: Step 3

Implement a Data Analytics / MLOps pipeline that automates storage, processing, and ingestion of data from one or more sources, based on your chosen dataset and use case.
This section of the log report should include a detailed discussion of the following **development** stages of the data analytics pipeline:

- Data Ingestion
  - Explain **how your pipeline collects and stores data** from one or more sources.
    - e.g., If your data comes from a CSV file, how does your pipeline read it and save it in a database for later use?
    - e.g., Have you used **ETL or ELT**? Why?
    - Are there any differences between the ingested/stored data and the initial data source used for ingestion? If so, why?
- Data Pre-processing
  - Explain **how your pipeline prepares the data** for Machine Learning.
    - e.g., Are there any **outliers** or **missing values** within each of your data variables? If yes,
      - how do you handle them?
        - Is the outlier likely to be correct based on what you know about the data?
        - Could the outlier be an example of a data entry error?
        - What does distribution of your data reveal?
    - e.g., Is your target variable balanced?? If no,
      - how do you handle them?
      - how this effect your model and evaluation pipeline?
- Model Development
  - Explain **how your pipeline trains a Machine Learning model** using the processed data.
    - e.g., How do you split data for **training and testing**? Justify.
    - e.g., Which **supervised learning model** you used? Why?
- Model Deployment
  - Explain **how your trained model is stored and made available for use**.
    - e.g., how are the trained ML model versions stored for later uses? (hint: serialisation?)
    - e.g., how might the stored models be accessible not only on a local machine, but also via a network? (hint: containers, like docker?)
- Model Monitoring
  - Explain how you check your **model's performance** after deployment.
    - e.g., If the model is predicting sales, how do you track whether its predictions remain accurate over time?

For each stage, you should aim to discuss:

- The What
  - Explain what this stage does.?
    - e.g., what does "data ingestion" actually mean?
  - What are the key components, systems and concepts that make up this stage?
    - e.g., what tools and did you use for the data ingestion process?
  - What are the input dependencies to this stage?
    - e.g., the data ingestion stage will likely require data
      - How has the data been passed into this stage of the pipeline?

        - Were there any requirements as to how that data needs to be represented/structured as input? If so, why?
    - What are the expected outputs of this stage?
        - e.g., once the data ingestion stage is completed, what will the later stages actually "get" out of this stage?

- The Why
    - Why is this stage important?
        - e.g. why was the data pre-processing stage needed?
    - Why are the key components, systems and concepts that make up this stage important?
        - e.g. what does each tool or technique contribute to the task being undertaken?
- The How
    - How was the stage implemented?

Tip

The design and development process of real-world systems are often iterative, this means that you are likely to benefit from refining your process throughout the development lifecycle. Therefore, this task should not be seen as a static "tick-box" document, instead, it should be reviewed throughout your progress with the module assessment and iteratively refined as you gain a better understanding of how to appropriately tackle the problem. Importantly, later more refined versions of your plan can then be compared against earlier versions

Marking

Marks will be awarded within this task based on:

- **Successful implementation** of an **appropriately documented** and **well-evidenced data analytics pipeline** accompanied by a sufficiently **detailed discussion** of the **development**, **setup,** and **configuration** process for said pipeline.
    - Try to show your understanding of **what happens at each stage of the process**?
    - Want **good marks?**
        - Then you should focus on showing your growing understanding of the 'whats', 'whys', 'whos' and 'hows' involved in the implementation of this pipeline
        - Have you presented high-quality, easy-to-follow, evidence of implementations in the form of diagrams, figures, tables, etc?
        - How well have you discussed the process of iteratively refining your pipeline as you progress throughout the semester?
        - Have you presented **high-quality**, **easy-to-follow**, **evidence** of your **proposed storage strategy** in the form of diagrams, figures, tables, etc?
        - How well have you discussed the **benefits** and **drawbacks** of your storage strategy?
        - The **logical** and **physical** structures imposed upon the data will always introduce some **trade-offs**, **what** are they? **why** are they acceptable for your intended use of the data?
- A **reflective discussion** of the process undertaken including:
    - How does your current pipeline compare to your initial plans?
    - How would you improve upon the structure of the pipeline in the future?
    - How you managed this project within your group?